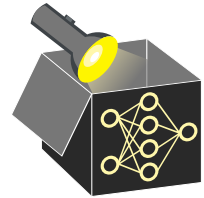
Interpretable Machine Learning

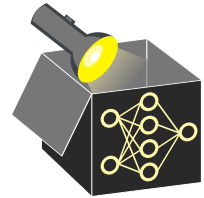
Marginal Effects

LEARNING GOALS



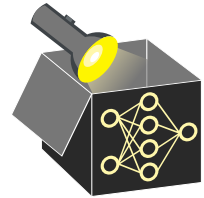
- Understand why parameter-based interpretations are not always possible for parametric models.
- Learn how marginal effects can be used to interpret such models.
- Recognize the drawbacks of traditional marginal effects.
- Discover model-agnostic applicability of forward marginal effects (FMEs).
- Introduce unified FMEs for all feature types.
- Learn about the Non-Linearity Measure (NLM) to assess deviation from linearity.
- Explore Conditional Average Marginal Effects (cAMEs) for regional interpretations.

INTRODUCTION



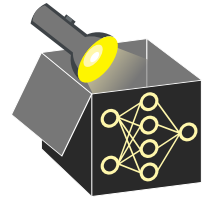
- **Motivation**:
  - Machine learning models are powerful but often lack interpretability.
  - Need for methods that provide interpretable insights similar to traditional statistical models.
- **Challenges**:
  - Complex, non-linear prediction functions.
  - Diverse feature types (continuous, categorical, mixed).
  - Interactions and higher-order terms complicate interpretations.
- **Objective**:
  - Develop model-agnostic methods to interpret machine learning models using Marginal Effects.
  - Introduce a unified approach for both continuous and categorical features.

INTERPRETATION OF SIMPLE MODELS



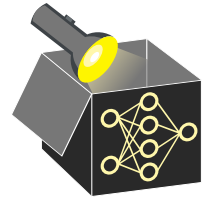
- **Linear Models**:
  - Change in x_j by Δx_j results in change in y by $\Delta y = \Delta x_j \cdot \theta_j$
  - Model equation:
$$y = \theta_0 + \theta_1 x_1 + \cdots + \theta_p x_p + \epsilon$$
  - Default interpretations correspond to $\Delta x_j = 1$, i.e., $\Delta y = \theta_j$
  - Assumes "ceteris paribus" (all other features held constant)
- **Non-Linear Models with Interactions**:
  - For models with higher-order or interaction terms, single coefficients are not sufficient:
$$y = \theta_0 + \theta_1 x_1^2 + \theta_2 x_2^2 + \theta_{1,2} x_1 x_2 + \epsilon$$
  - Marginal effect of x_1 varies with different values of x_2 (and vice versa)
  - Interactions depend on the values of other features

MARGINAL EFFECTS



- **Definition**:
  - Marginal Effects (MEs) measure the change in the predicted outcome due to a change in a feature.
- **Types of MEs**:
  - **Derivative Marginal Effects (DMEs)**: Use derivatives; suitable for continuous, differentiable functions.
  - **Forward Marginal Effects (FMEs)**: Use finite differences; applicable to non-differentiable functions.
- **Limitations of DMEs**:
  - Not suitable for non-differentiable or piecewise constant functions (e.g., decision trees).
  - Interpretation can be misleading for non-linear models.

MARGINAL EFFECTS FOR CONTINUOUS FEATURES



- **Derivative Marginal Effect (DME)**:

$$\text{DME}_j(\mathbf{x}) = \frac{\partial \widehat{f}(\mathbf{x})}{\partial x_j} \approx \frac{\widehat{f}(\mathbf{x} + h_j \mathbf{e}_j) - \widehat{f}(\mathbf{x} - h_j \mathbf{e}_j)}{2h_j}$$

  where $\mathbf{e}_j$ is the unit vector in the j-th direction.

- **Forward Marginal Effect (FME)**:

$$\text{FME}_j(\mathbf{x}, h_j) = \widehat{f}(\mathbf{x}_1, \ldots, x_j + h_j, \ldots, \mathbf{x}_p) - \widehat{f}(\mathbf{x})$$

- **Example Model**:

$$y = \theta_0 + \theta_1 x_1^2 + \theta_2 x_2^2 + \theta_{1,2} x_1 x_2 + \epsilon$$

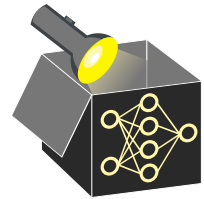
  - Derivative ME of x_1:

$$\text{DME}_1(\mathbf{x}) = 2\theta_1 x_1 + \theta_{1,2} x_2$$

  - Forward ME of x_1 with step size h_1:

$$\text{FME}_1(\mathbf{x}, h_1) = \theta_1 \left((x_1 + h_1)^2 - x_1^2 \right) + \theta_{1,2} x_2 h_1$$

MARGINAL EFFECTS FOR CATEGORICAL FEATURES

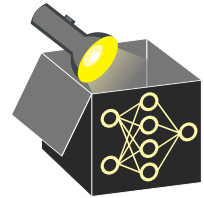


- **Traditional Approach**:
  - Compute the change in prediction when the feature value changes from a reference category to another category.
  - For each observation, set the categorical feature to the reference category and record the change in prediction when changing it to every other category.
- **Unified FME Definition for Categorical Features**:

$$\text{FME}_{\mathbf{x},c_j} = \widehat{f}(c_j, \mathbf{x}_{-j}) - \widehat{f}(\mathbf{x})$$

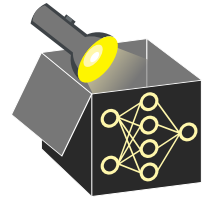
- **Advantages**:
  - Removes ambiguity by treating categorical features similarly to continuous features.
  - Provides an exact change in prediction for a change in category.

LIMITATIONS OF AMES IN NON-PARAMETRIC MODELS



- **Average Marginal Effects (AMEs)**:
  - Widely used in logistic regression to interpret effects on the probability scale.
  - Provide an average effect across all observations.
- **Variations**:
  - Marginal Effects at the Mean (MEM).
  - Marginal Effects at Representative Values (MER).
- **Issue with Non-Parametric Models**:
  - AMEs assume a consistent effect across the feature space.
  - Non-parametric models (e.g., random forests, neural networks) can model complex, non-linear relationships.
  - Averaging effects can obscure important heterogeneities.

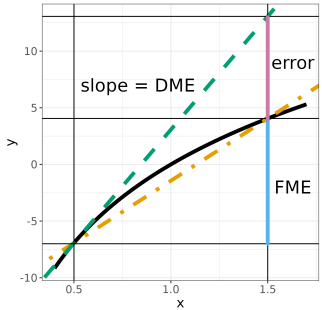
EXAMPLE: AMES IN LOGISTIC REGRESSION VS. NON-PARAMETRIC MODEL

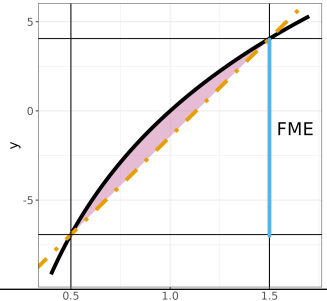


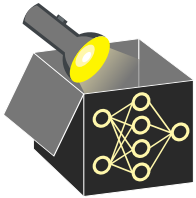
- **Logistic Regression Model**:
  - Model: $\text{logit}(P(Y = 1)) = \beta_0 + \beta_1 X$
  - AME provides average change in probability for a unit change in X.
  - Transformation is monotonic, so averaging makes sense.
- **Non-Parametric Model (e.g., Decision Tree)**:
  - Model can capture non-linear and interaction effects.
  - Effect of X on Y can vary greatly across different regions.
  - **AME may not reflect any individual's experience**.
- **Conclusion**:
  - Need for local or regional methods like FMEs and cAMEs to capture true effects.

FORWARD DIFFERENCE VS. DERIVATIVE

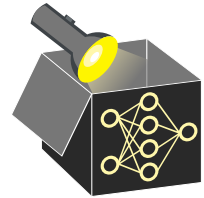
- Derivatives of non-linear functions vary significantly at different points $\rightarrow$ DME may not be reliable.
- FME captures the actual change in the prediction due to a finite change in the feature.
- FME is more suitable for non-linear and non-differentiable models.
- Both methods may lose information about the function's behavior along the change.
- **Visualization**:
  - DME corresponds to the slope of the tangent at a point.
  - FME corresponds to the slope of the secant between two points.







ADDITIVE RECOVERY PROPERTY



- Both DME and FME recover only terms involving the feature(s) of interest.
- **Example**:

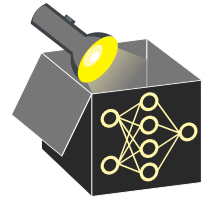
$$\widehat{f}(\mathbf{x}) = ax_1 + bx_2$$

Then:

$$\mathrm{DME}_1(\mathbf{x}) = a$$
$$\mathrm{FME}_1(\mathbf{x}, h_1) = ah_1$$

- Effects of other features linked additively are removed.
- This property allows focusing on the direct effect of the feature.

UNIFIED DEFINITION OF FMES



- **For Continuous Features**:

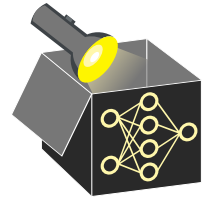
$$\text{FME}_{\mathbf{x},\boldsymbol{h}_S} = \widehat{f}(\mathbf{x}_S + \boldsymbol{h}_S, \mathbf{x}_{-S}) - \widehat{f}(\mathbf{x})$$

- **For Categorical Features**:

$$\text{FME}_{\mathbf{x},c_j} = \widehat{f}(c_j, \mathbf{x}_{-j}) - \widehat{f}(\mathbf{x})$$

- **Mixed-Type Features**:
  - Combine changes in both continuous and categorical features.
- **Advantages**:
  - Provides exact change in prediction.
  - Applicable to any model type.
  - Removes ambiguity from marginal effects definitions.

NON-LINEARITY MEASURE (NLM)

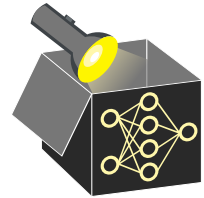


- **Purpose**:
  - Quantify deviation from linearity along the path of the FME.
- **Computation**:
  - Compare the prediction function to the linear secant.
  - Use the normalized integral of squared deviations.
- **Definition**:

$$\mathrm{NLM}_{\mathbf{x},\boldsymbol{h}_S} = 1 - \frac{\int_0^1 [\widehat{f}(\gamma(t)) - g_{\mathbf{x},\boldsymbol{h}_S}(t)]^2 \|\gamma'(t)\|_2 dt}{\int_0^1 [\widehat{f}(\gamma(t)) - \bar{\widehat{f}}]^2 \|\gamma'(t)\|_2 dt}$$

- **Interpretation**:
  - $\mathrm{NLM} \in (-\infty, 1]$.
  - $\mathrm{NLM} = 1$: Perfect linearity.
  - $\mathrm{NLM} \leq 0$: Indicates strong non-linearity.

COMPUTATION AND INTERPRETATION OF NLM



- **Linear Reference Function**:

$$g_{\mathbf{x},\boldsymbol{h}_S}(t) = \widehat{f}(\mathbf{x}) + t \cdot \text{FME}_{\mathbf{x},\boldsymbol{h}_S}$$

- **Path Parameterization**:

$$\gamma(t) = \mathbf{x} + t \cdot \boldsymbol{h}_S$$

- **Line Integrals**:

$$(\text{I}) = \int_0^1 [\widehat{f}(\gamma(t)) - g_{\mathbf{x},\boldsymbol{h}_S}(t)]^2 \|\boldsymbol{h}_S\|_2 dt$$

$$(\text{II}) = \int_0^1 [\widehat{f}(\gamma(t)) - \bar{\widehat{f}}]^2 \|\boldsymbol{h}_S\|_2 dt$$

- **Advantages**:
  - Provides a quantitative measure of non-linearity.
  - Applicable to univariate and multivariate feature changes.
  - Helps assess the reliability of linear approximations.

VISUALIZATION OF NLM

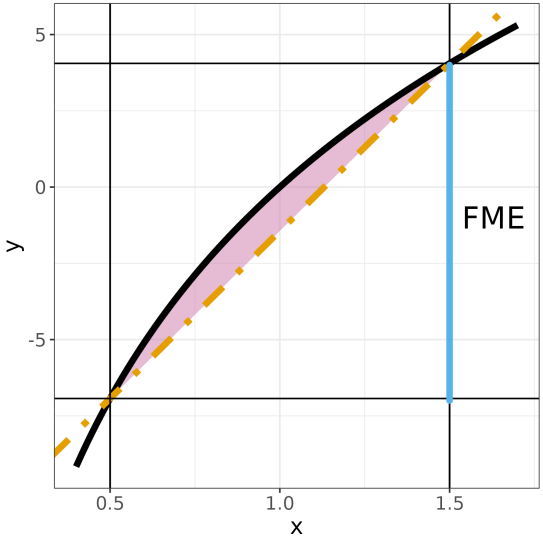
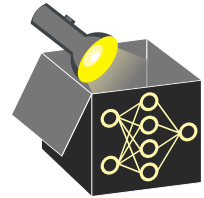


Figure: Comparison between the prediction function and the linear secant. The purple area represents the deviation quantified by the NLM.

SELECTING STEP SIZES AND TRUST REGIONS



- **Step Size Selection**:
  - Based on domain knowledge or natural units.
  - Use statistical measures (e.g., standard deviation, IQR).
  - Contextual information dictates meaningful step sizes.
- **Local Linear Trust Region (LLTR)**:
  - Identify regions where NLM indicates approximate linearity (e.g., NLM ≥ 0.9).
  - Allows for reliable interpretation of FMEs.
  - Step sizes within LLTR ensure linear approximation is valid.

LOCAL LINEAR TRUST REGION (LLTR) VISUALIZATION

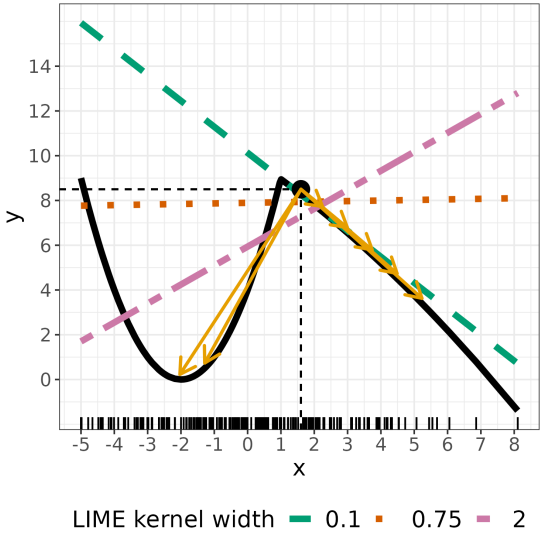
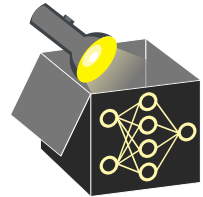
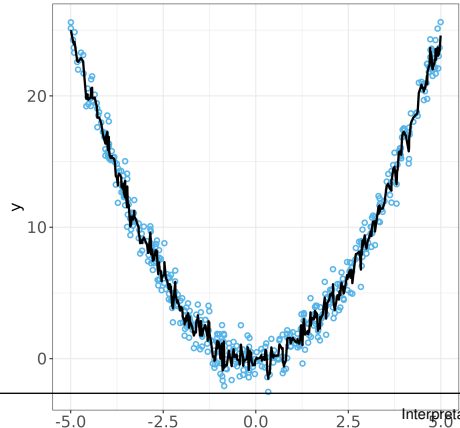


Figure: LLTR using FMEs compared to LIME explanations. The orange arrows represent steps where NLM indicates linearity.

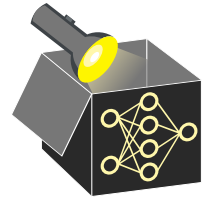
MODEL EXTRAPOLATION AND STEP SIZES



- **Model Extrapolation Risks**:
  - Large step sizes may lead to extrapolation beyond the training data.
  - Predictions in extrapolated regions may be unreliable.
- **Mitigation**:
  - Limit step sizes to ranges within the training data.
  - Identify and exclude extrapolation points (EPs) from analysis.
- **Visualization**:



CONDITIONAL AVERAGE MARGINAL EFFECTS (CAMES)

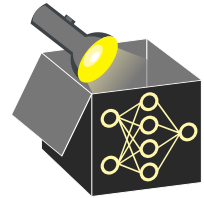


- **Purpose**:
  - Estimate expected FMEs within specific subgroups or regions.
  - Capture regional effects in non-linear models.
- **Definition**:

$$\text{cAME}_{\mathcal{D}_{[\,]},\boldsymbol{h}_S} = \frac{1}{n_{[\,]}} \sum_{i\,:\,\mathbf{x}^{(i)} \in \mathcal{D}_{[\,]}} \left[\widehat{f}\left(\mathbf{x}_S^{(i)} + \boldsymbol{h}_S, \mathbf{x}_{-S}^{(i)}\right) - \widehat{f}\left(\mathbf{x}^{(i)}\right) \right]$$

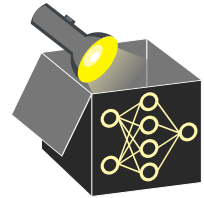
- **Subgroups**:
  - Defined based on feature values or characteristics.
  - Aim to have homogeneous effects within subgroups.

DESIDERATA FOR FINDING SUBGROUPS



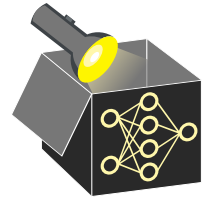
- **Within-Group Effect Homogeneity**:
  - Minimize FME variance within subgroups.
- **Between-Group Effect Heterogeneity**:
  - Maximize differences in cAMEs between subgroups.
- **Full Segmentation**:
  - Entire dataset is partitioned into subgroups.
- **Non-Congruence**:
  - Subgroups do not overlap.
- **Confidence and Stability**:
  - Prefer larger, stable subgroups.
  - Subgroups should be consistent across data variations.

ESTIMATION USING DECISION TREES



- **Recursive Partitioning (RP)**:
  - Use decision trees to partition the feature space.
  - Find subgroups with homogeneous FMEs.
- **Conditional Inference Trees (CTREE)**:
  - Use statistical tests to find significant splits.
  - Help stabilize the splits and improve subgroup reliability.
- **Advantages**:
  - Automatically identify meaningful subgroups.
  - Handle complex interactions and non-linearities.

CONFIDENCE INTERVALS FOR CAME AND CANLM



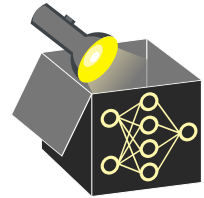
- **Purpose**:
  - Provide uncertainty estimates for cAMEs and cANLMs.
- **Computation**:

$$\mathrm{CI}_{\mathrm{cAME},\,1-\alpha} = \left[\mathrm{cAME}_{\mathcal{D}_{[\,]},\boldsymbol{h}_S} \pm t_{1-\frac{\alpha}{2},n_{[\,]}-1}\frac{\mathrm{SD}_{\mathrm{FME},[\,]}}{\sqrt{n_{[\,]}}}\right]$$

$$\mathrm{CI}_{\mathrm{cANLM},\,1-\alpha} = \left[\mathrm{cANLM}_{\mathcal{D}_{[\,]},\boldsymbol{h}_S} \pm t_{1-\frac{\alpha}{2},n_{[\,]}-1}\frac{\mathrm{SD}_{\mathrm{NLM},[\,]}}{\sqrt{n_{[\,]}}}\right]$$

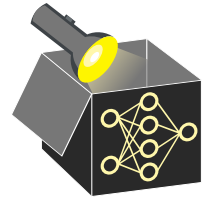
- **Interpretation**:
  - Narrow intervals indicate higher confidence.
  - Larger subgroups provide more reliable estimates.

RELATION TO ICE AND PD



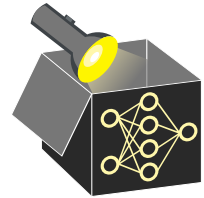
- **Individual Conditional Expectation (ICE)**:
  - Visualizes predictions for an observation across a range of feature values.
  - FME corresponds to vertical differences between points on an ICE curve.
- **Partial Dependence (PD)**:
  - Shows average predictions across a range of feature values.
  - AME is equivalent to vertical differences on PD for linear models.
- **Advantages of FMEs**:
  - Provide exact change in prediction.
  - Applicable to high-dimensional feature changes.
  - Quantifiable and not limited to visual interpretation.

COMPARISON TO LIME



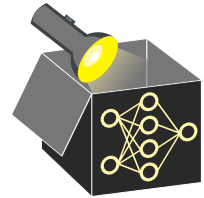
- **LIME (Local Interpretable Model-Agnostic Explanations)**:
  - Trains a local surrogate model around an observation.
  - Interpretation depends on surrogate model parameters.
- **Differences with FMEs**:
  - FMEs directly compute changes without surrogates.
  - FMEs are deterministic with fixed parameters.
  - LIME explanations can vary due to randomness and parameter choices.
- **Advantages of FMEs**:
  - More stable and interpretable.
  - NLM provides measure of linearity and trust in the explanation.

RELATION TO SENSITIVITY ANALYSIS



- **Purpose of Sensitivity Analysis (SA)**:
  - Determine how uncertainty in model output can be attributed to inputs.
- **Methods in SA**:
  - Use finite differences to estimate local sensitivities.
  - Elementary effects correspond to univariate forward difference quotients.
- **Relation to FMEs**:
  - FMEs can be seen as a form of sensitivity analysis.
  - Provide insight into feature importance and interactions.

CONCLUSION AND FUTURE WORK



- **Key Takeaways**:
  - FMEs provide exact changes in predictions for any model type.
  - NLM helps assess the linearity of effects along feature changes.
  - cAMEs enable regional interpretations, capturing heterogeneity.
  - Unified approach removes ambiguity from marginal effects.
- **Future Directions**:
  - Enhance methods for detecting model extrapolations.
  - Explore stabilization techniques for subgroup discovery.
  - Investigate applications of FMEs in feature importance and other domains.
  - Integrate FMEs with other interpretability methods.